



ABSTRACTION



Student Management System (Week 1)

The Student Management System is designed to simplify the management of student records by providing an easy way to store and maintain student information. The system abstracts the real-world process of managing students by representing each student as an object containing only the essential details such as **Student ID**, **Name**, and **Department**.

Users interact with the system through simple menu-driven operations like adding, viewing, updating, and deleting student records, without needing to understand how the data is stored internally. The application uses an **ArrayList** to maintain the collection of student objects, while **encapsulation** ensures that student data is protected and accessed only through appropriate methods.

By hiding the implementation details and exposing only the necessary operations, the system provides a simple and user-friendly interface. This abstraction improves code organization, reduces complexity, and makes the application easier to maintain and extend.

As a result, the Student Management System serves as an efficient console-based application for managing student information accurately and effectively.

